SOLAR RADIATION ANALYSIS & PREDICTION

STAC WARS · Problem Statement 2

A data analytics and machine-learning study of solar radiation, environmental conditions, and daily solar activity.

Project focus	Exploratory visualization · Evidence-based inference · Daily aggregation · Regression prediction
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn
Dataset	Voyage-123 solar observation dataset (source referenced in the problem statement)
Prepared for	STAC WARS — PS 2: Solar Radiation Analysis and Prediction

Important data-integrity note: The problem statement was available, but the actual Voyage-123 dataset was not attached with it. Therefore, this report provides the complete analysis framework, reproducible code, evaluation plan, and clearly marked result fields. Dataset-dependent plots, numerical findings, top dates, and model scores must be generated from the real CSV before submission. No values have been fabricated.

May the Code Be With You

Executive Summary

This project investigates solar radiation recorded by the Voyage-123 observation probe and examines how it varies with time and environmental conditions. The analysis is organized around the four requirements in PS 2: (1) data visualization, (2) drawing evidence-based inferences, (3) day-wise solar activity analysis using Pandas and NumPy, and (4) a bonus regression model to predict solar radiation.

The workflow begins with data validation and datetime conversion, followed by exploratory visualization, daily aggregation, and supervised regression. The model is evaluated on a held-out test set using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Feature importance or model coefficients are used to discuss which inputs contribute most to prediction.

The final scientific conclusions must be based on the actual dataset outputs. This document intentionally distinguishes planned analyses from observed findings so the report remains defensible during judging.

Problem statement objectives

• Visualize how solar radiation changes over time.

• Explore radiation relationships with at least two environmental parameters.

• Use plots to identify trends, patterns, and potential outliers.

• Determine the time of day when radiation is generally highest and examine relationships among radiation, temperature, and humidity.

• Convert date/time fields to a proper datetime column and calculate daily maximum and average radiation.

• Find the date of the highest recorded radiation, its timestamp, associated temperature, humidity, and wind speed, and the elapsed time from sunrise to the peak.

• Identify the five dates with the highest average radiation.

• Train a regression model, evaluate it using RMSE and R^2, and explain useful predictive features.

1. Introduction

Solar radiation is the solar energy received at the Earth's surface. It changes throughout the day due to the sun's position and may also vary with cloud cover, atmospheric moisture, temperature, wind, and other local conditions. Understanding these patterns can support solar-energy planning, performance monitoring, and operational decisions.

The Voyage-123 dataset is described in the assignment as containing solar observations linked to radiation, temperature, humidity, wind, sunrise, and sunset. The project treats solar radiation as the primary variable of interest and uses the available time and environmental variables to explain observed variation and build a predictive baseline.

1.1 Project scope

The scope is limited to the columns and time period available in the supplied dataset. The report does not assume that a particular relationship is causal. Correlation and model importance indicate association or predictive usefulness, not proof that one variable causes changes in solar radiation.

1.2 Expected deliverables

• Python notebook or .py source files committed to the GitHub repository.

• Time-series and relationship plots with readable labels, units where provided, and captions.

• A written inference section supported by computed statistics and visual evidence.

• Daily summary table, peak-radiation record, sunrise-to-peak calculation, and top-five daily averages.

• A trained regression model with reproducible train/test split and RMSE/R^2 results.

• A concise discussion of limitations and next steps.

2. Dataset Understanding and Preparation

Dataset source: Voyage-123 solar observation data, linked from the original problem statement. The source file itself must be downloaded and included or linked in the repository so the analysis can be reproduced.

2.1 Required fields

Field / field group	Purpose in analysis	Validation
Date and Time	Create a single timestamp; sort records and derive date/hour features.	Parse successfully; inspect missing or invalid timestamps.
Solar radiation	Target variable for visualization, aggregation, and prediction.	Check missing values, units, impossible values, and outliers.
Temperature	Compare thermal conditions with radiation and use as a predictor.	Check numeric type and missingness.
Humidity	Explore moisture-related association with radiation.	Check numeric type and valid range if units are known.
Wind speed	Inspect relationship and report conditions at peak radiation.	Check numeric type and missingness.
Sunrise / sunset	Compare peak time with daylight boundaries.	Parse according to actual file format and local time convention.

2.2 Data-cleaning protocol

• Read the source file and inspect its shape, column names, data types, and sample rows.

• Normalize column names carefully; do not rename columns until the actual file schema is inspected.

• Convert the supplied Date and Time fields into a timestamp using the format present in the data.

• Convert radiation and environmental measurements to numeric values; flag conversion failures rather than silently replacing them.

• Quantify missing values and duplicate records. Remove a row only when there is a clear reason and record the rule in the notebook.

• Sort observations chronologically and check the sampling interval, duplicate timestamps, and suspicious gaps.

• Keep a record of cleaning decisions so another person can reproduce the results.

Data-quality summary to complete after loading the dataset: number of rows = [computed value]; number of columns = [computed value]; date range = [computed range]; missing-value summary = [computed table]; duplicate count = [computed value].

3. Methodology

3.1 Exploratory data analysis

Exploratory data analysis (EDA) is used to understand distributions and relationships before selecting a model. Begin with descriptive statistics for radiation and each candidate environmental variable. Use line plots for temporal patterns, scatter plots for pairwise relationships, and a correlation matrix as a compact overview. A plot should have a meaningful title, labelled axes, and a short interpretation tied to visible evidence.

3.2 Visualization plan

Figure	Recommended plot	Question answered
1	Solar radiation vs timestamp (line plot)	How does radiation vary over time? Are there daily cycles or abrupt spikes?

Figure	Recommended plot	Question answered
2	Radiation vs temperature (scatter plot)	Is there an apparent association, and are there outliers?
3	Radiation vs humidity (scatter plot)	Does radiation tend to differ across humidity levels?
4	Radiation vs wind speed (scatter plot)	Is wind speed associated with radiation in this dataset?
5 (bonus)	Hour-of-day radiation profile or correlation heatmap	What additional temporal or multivariate pattern appears?

3.3 Inference strategy

Every inference should follow a claim–evidence–interpretation pattern. First state the pattern, then quote the supporting statistic or chart feature, and finally explain what it may mean without claiming causation. For example: “Radiation is generally highest around [computed hour range], as shown by [plot or hourly summary]. This is consistent with the daily solar cycle in this dataset.” Replace the bracketed text only after the results are computed.

3.4 Day-wise solar activity report

Create a date-only grouping from the parsed timestamp. For each date, calculate maximum radiation and mean radiation. Identify the row corresponding to the overall maximum radiation and report its timestamp, temperature, humidity, and wind speed. Compare the peak timestamp with that date's sunrise and report the elapsed time. Finally, sort the daily mean radiation table in descending order and return the first five dates.

3.5 Machine-learning approach

Treat solar radiation as a continuous target and use regression. A transparent baseline such as Random Forest Regressor is suitable for nonlinear relationships among environmental and temporal variables. Use a fixed random seed for reproducibility, hold out a test set, and calculate RMSE and R^2. If the dataset is time-ordered and the purpose is to predict future observations, a chronological split is more realistic than a random split; document which strategy is used.

Avoid target leakage: do not include the radiation value itself, a transformation derived from the target, or any future information that would not be available at prediction time among the inputs. If sunrise/sunset are text fields, parse them and represent them appropriately before modelling.

4. Results and Interpretation

This section is deliberately structured for the real computed outputs. Fill every bracketed field after running the notebook against the official dataset. The actual data file was not included with the problem statement attachment, so numerical results cannot responsibly be reported here.

4.1 Dataset profile

Metric	Computed result
Number of observations	[Insert row count]
Observation date range	[Insert start and end date]
Radiation mean / median	[Insert computed values and units]
Radiation minimum / maximum	[Insert computed values and units]
Missing values / duplicate timestamps	[Insert counts]

4.2 Visualization findings

• Temporal trend: [Describe the pattern visible in Figure 1, including the time scale and any notable spikes or gaps.]

• Radiation and temperature: [Describe direction, strength, spread, and any exceptions visible in Figure 2.]

• Radiation and humidity: [Describe the pattern in Figure 3; avoid assuming causality.]

• Radiation and wind speed: [Describe the pattern in Figure 4.]

• Bonus insight: [Explain the additional pattern revealed by the hourly profile or correlation heatmap.]

4.3 Answers to the assignment's inference questions

Question	Evidence-based answer to complete
When is radiation generally highest?	[Insert peak hour/range from hourly aggregation and cite the figure.]
Which environmental parameter has the strongest relationship with radiation?	[Insert strongest absolute correlation or justified visual finding; state the metric used.]
What pattern links radiation, temperature, and humidity?	[Summarize the observed scatter plots/correlation values; mention any exceptions.]
Are there unusual observations or outliers?	[List timestamp(s) and values, then describe plausible explanations as hypotheses only.]

4.4 Daily solar activity results

Required output	Result
Date with highest recorded radiation	[Insert date]
Peak timestamp and radiation value	[Insert timestamp and value]
Temperature at peak	[Insert value and unit]
Humidity at peak	[Insert value and unit]
Wind speed at peak	[Insert value and unit]
Sunrise time and time from sunrise to peak	[Insert sunrise time and elapsed duration]
Top 5 days by average radiation	[Insert dates and mean radiation values in a ranked table]

4.5 Prediction results

Item	Result
Model	[Insert model and hyperparameters]
Features used	[List features after leakage review]
Train/test split	[Insert split strategy, ratio, and random seed or cutoff date]
RMSE	[Insert test RMSE; include radiation units]
R^2	[Insert test R^2]
Most useful features	[Insert feature importance or coefficient-based explanation]

Interpretation guidance: lower RMSE indicates smaller typical prediction errors in the target's units. R^2 describes the fraction of test-set variance explained relative to a mean-prediction baseline; it can be negative if the model performs worse than that baseline. Do not describe a model as “accurate” based on R^2 alone—inspect the error scale, actual-versus-predicted plot, and residuals.

5. Reproducible Python Implementation

The following reference script is designed to be adapted after checking the real dataset's column names and file format. It expects a CSV containing the assignment's Date and Time fields plus radiation, temperature, humidity, wind speed, and (if available) sunrise/sunset. Update the configuration section to match the exact source schema. The code does not replace a careful review of missing values, units, or time formats.

```
# Install once if needed:
# pip install pandas numpy matplotlib seaborn scikit-learn

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.inspection import permutation_importance

# 1) Load and inspect the source data
df = pd.read_csv("voyage123_solar_data.csv")
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
print(df.head())
print(df.isna().sum())

# 2) Configure these names to match the actual CSV exactly
DATE_COL = "Date"
TIME_COL = "Time"
RAD_COL = "Solar Radiation"
TEMP_COL = "Temperature"
HUM_COL = "Humidity"
WIND_COL = "Wind Speed"

# If column names differ, change the constants above.
df["timestamp"] = pd.to_datetime(
    df[DATE_COL].astype(str) + " " + df[TIME_COL].astype(str),
    errors="coerce"
)
df = df.dropna(subset=["timestamp", RAD_COL]).copy()
for col in [RAD_COL, TEMP_COL, HUM_COL, WIND_COL]:
    df[col] = pd.to_numeric(df[col], errors="coerce")
df = df.sort_values("timestamp")
df["date"] = df["timestamp"].dt.date
df["hour"] = df["timestamp"].dt.hour

print("Cleaned shape:", df.shape)
print("Date range:", df["timestamp"].min(), "to", df["timestamp"].max())

# 3) Visualizations
plt.figure(figsize=(12, 4))
plt.plot(df["timestamp"], df[RAD_COL], linewidth=0.8)
plt.title("Solar Radiation Over Time")
plt.xlabel("Timestamp")
plt.ylabel("Solar radiation (dataset units)")
plt.tight_layout()
plt.savefig("01_radiation_over_time.png", dpi=200)
plt.show()

for col, filename in [
    (TEMP_COL, "02_radiation_vs_temperature.png"),
    (HUM_COL, "03_radiation_vs_humidity.png"),
    (WIND_COL, "04_radiation_vs_wind_speed.png"),
]:
    if col in df.columns:
        plt.figure(figsize=(6, 4))
        sns.scatterplot(data=df, x=col, y=RAD_COL, alpha=0.45, s=18)
        plt.title(f"Solar Radiation vs {col}")
        plt.tight_layout()
        plt.savefig(filename, dpi=200)
        plt.show()

plt.figure(figsize=(8, 4))
df.groupby("hour")[RAD_COL].mean().plot(marker="o")
plt.title("Mean Solar Radiation by Hour of Day")
plt.xlabel("Hour of day")
plt.ylabel("Mean solar radiation (dataset units)")
plt.tight_layout()
plt.savefig("05_hourly_radiation_profile.png", dpi=200)
plt.show()

numeric_cols = df.select_dtypes(include=np.number).columns
plt.figure(figsize=(8, 6))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap="coolwarm",
            center=0, fmt=".2f")
plt.title("Correlation Heatmap")
plt.tight_layout()
```

```
plt.savefig("06_correlation_heatmap.png", dpi=200)
plt.show()

# 4) Evidence for inferences
print("Descriptive statistics:")
print(df[[RAD_COL, TEMP_COL, HUM_COL, WIND_COL]].describe())
print("Correlations with radiation:")
print(df[numeric_cols].corr()[RAD_COL].sort_values(ascending=False))
hourly_mean = df.groupby("hour")[RAD_COL].mean().sort_values(ascending=False)
print("Hours ranked by mean radiation:")
print(hourly_mean)

# 5) Daily report
daily = df.groupby("date")[RAD_COL].agg(
    daily_max_radiation="max",
    daily_average_radiation="mean"
).reset_index()
print("Daily report:")
print(daily.sort_values("date"))
print("Top 5 days by average radiation:")
print(daily.nlargest(5, "daily_average_radiation"))

peak_idx = df[RAD_COL].idxmax()
peak = df.loc[peak_idx]
print("Overall peak observation:")
print(peak[["timestamp", RAD_COL, TEMP_COL, HUM_COL, WIND_COL]])

# Sunrise calculation depends on the source's actual time format.
# If a sunrise column exists and can be parsed, calculate the difference
# between peak timestamp and that date's sunrise after validating timezone
# and date conventions. Do not compare time-only strings without parsing.

# 6) Regression model: choose only columns available at prediction time.
feature_candidates = [TEMP_COL, HUM_COL, WIND_COL, "hour"]
features = [c for c in feature_candidates if c in df.columns]
model_df = df[features + [RAD_COL]].dropna().copy()
X = model_df[features]
y = model_df[RAD_COL]

# Random split for a baseline. For future forecasting, use a chronological
# split instead and keep later timestamps entirely in the test set.
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
model = RandomForestRegressor(
    n_estimators=300, random_state=42, min_samples_leaf=2, n_jobs=-1
)
model.fit(X_train, y_train)
pred = model.predict(X_test)

rmse = np.sqrt(mean_squared_error(y_test, pred))
r2 = r2_score(y_test, pred)
print(f"Test RMSE: {rmse:.4f}")
print(f"Test R2: {r2:.4f}")

plt.figure(figsize=(5, 5))
plt.scatter(y_test, pred, alpha=0.5)
lims = [min(y_test.min(), pred.min()), max(y_test.max(), pred.max())]
plt.plot(lims, lims, "k--")
plt.xlabel("Actual radiation")
plt.ylabel("Predicted radiation")
plt.title("Actual vs Predicted Solar Radiation")
plt.tight_layout()
plt.savefig("07_actual_vs_predicted.png", dpi=200)
plt.show()

importance = pd.Series(
    model.feature_importances_, index=features
).sort_values(ascending=False)
print("Random Forest feature importance:")
print(importance)
```

6. Model Evaluation and Validation

6.1 Evaluation metrics

• **RMSE:** The square root of mean squared prediction error. It is expressed in the same units as solar radiation and penalizes larger errors more strongly.

• **R^2:** Measures test-set fit relative to predicting the test target mean. A value closer to 1 is generally better; 0 indicates performance similar to the mean baseline, and a negative value indicates worse performance than that baseline.

6.2 Validation checklist

• Keep the test set separate from model fitting and feature selection.

• Use a fixed random seed for a reproducible random split, or document the chronological cutoff for a time-based split.

• Compare against a simple baseline (for example, predicting the training-set mean).

• Inspect actual-versus-predicted values and residuals, not only a single metric.

• Check whether input features would genuinely be available when a prediction is made.

• If timestamps from the same day are highly related, consider a time-aware or day-wise split to reduce overly optimistic evaluation.

• Record package versions and model parameters in the notebook or README.

7. Discussion, Limitations, and Responsible Interpretation

7.1 Potential limitations

• The analysis is constrained by the time span, sampling frequency, measurement quality, and completeness of the supplied observations.

• Environmental associations may be confounded by unrecorded factors such as cloud cover, season, sensor orientation, or atmospheric conditions.

• A random train/test split can overstate performance when nearby timestamps are strongly correlated; chronological validation may better represent future prediction.

• Feature importance describes the model's use of features, not causal effects. Correlated predictors can share or distort importance.

• Outliers should be investigated, not automatically deleted; they may represent real events, sensor faults, or data-entry issues.

7.2 Improvements

• Validate on a later time period and compare random-split versus chronological performance.

• Compare Random Forest with a linear regression baseline and a gradient-boosting model if appropriate.

• Engineer cyclical hour features using sine and cosine, and add day-of-year features if the dataset spans enough time.

• Add uncertainty estimates or prediction intervals where feasible.

• Document units and use physical plausibility checks before interpreting unusual measurements.

8. Conclusion

The project provides a reproducible workflow for investigating solar radiation through visualization, evidence-based inference, daily aggregation, and regression. The central outputs are the daily radiation summary, the record of the maximum-radiation observation, the sunrise-to-peak interval, the top five days by mean radiation, and a test-set evaluation using RMSE and R^2.

A strong final submission should replace every bracketed result field with values calculated from the official Voyage-123 dataset, insert the generated figures with captions, and ensure that each conclusion points to a plot or statistic. Until the real dataset is processed, no specific peak hour, strongest environmental relationship, outlier explanation, top-five date, or predictive score can be asserted.

The final repository should include the notebook/script, source-data instructions, generated figures, and this report. A clean README should explain how to install dependencies and reproduce the results.

9. Repository and Submission Checklist

- ■ Include the analysis notebook (.ipynb) and/or Python script (.py).
- ■ Document where the official dataset came from and how to place it locally; do not commit restricted data.
- ■ Include generated figures in a `figures/` directory.
- ■ Include a `requirements.txt` with the packages used.
- ■ Add a README with project objective, dataset fields, execution steps, key results, and limitations.
- ■ Ensure the report contains the plots, inferences, daily report, and RMSE/R^2 scores required by PS 2.
- ■ Verify all numerical claims against code output before presenting to judges.
- ■ Remove all remaining bracketed placeholders before final submission.

Appendix A — Suggested Final Report Figure Captions

- **Figure 1.** Solar radiation over time. Discuss broad trends, periodicity, missing intervals, and unusual peaks.
- **Figure 2.** Solar radiation versus temperature. Describe the direction and spread of the relationship.
- **Figure 3.** Solar radiation versus humidity. Describe the observed association and any exceptions.
- **Figure 4.** Solar radiation versus wind speed. Explain whether a visible pattern exists.
- **Figure 5.** Mean solar radiation by hour of day. Identify the hour or hour range with the highest average.
- **Figure 6.** Correlation heatmap. Report the strongest relationship with radiation and clarify that correlation is not causation.
- **Figure 7.** Actual versus predicted radiation. Discuss alignment with the diagonal and any systematic errors.

Appendix B — Final Checks Before Judging

Judges are likely to value correctness, reproducibility, readable visualizations, a clear connection between evidence and claims, and an honest evaluation of the model. The report should therefore present the results before the interpretation, include units, explain the split strategy, and avoid unsupported claims. A polished report with transparent limitations is stronger than one containing attractive but unverifiable numbers.

Source basis: STAC WARS, PS 2 — Solar Radiation Analysis and Prediction (three-page problem statement supplied by the user). All workflow recommendations and code are implementation guidance; actual empirical findings must be computed from the official dataset.